

Kayra Bulbul

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EDUCATION

RMIT University

Melbourne, VIC

Bachelor of Computer Science, Minors in data science and Enterprise Systems development

Mar. 2024 – Nov 2027

EXPERIENCE

Peer Mentor

July 2026 – Present

RMIT University

Melbourne, VIC

- Mentor students in **Algorithms & Analysis** and **Further Programming**, helping develop understanding of data structures, algorithms, OOP and debugging techniques.
- Guide students through problem-solving approaches and debugging strategies rather than providing direct solutions, strengthening their ability to independently reason about technical problems.

ACHIEVEMENTS

Winner – MelbourneHack 2026 | *Python, FastAPI, PostgreSQL, ReactTS, Gemini 3.6 Flash*

August 2026

- Built an accessibility platform in a four-person team that transforms web content based on customisable accessibility needs; led backend and database development using FastAPI and PostgreSQL, integrating page transformation, persistence and frontend services.

PROJECTS

Distributed Chat System | *Go, WebSockets, Redis, Docker, Prometheus/Grafana*

August 2026 – Present

- Designed and built a **distributed real-time chat system** using WebSockets, supporting persistent client connections and communication across multiple server instances.
- Implemented **horizontal scaling** and **load balancing**, with Redis Pub/Sub enabling message propagation between users connected to different application servers.
- Engineered the system with an **infrastructure-focused architecture**, including containerised services with Docker, structured logging, metrics and observability, and load testing to evaluate behaviour under concurrent traffic.

Overworld Atlas | *React, TypeScript, Go, Docker, PostgreSQL*

July 2026 – Present

- Built a **Go backend** service for a Minecraft server I play on with my mates, exposing APIs for server status, authentication and application functionality.
- Implemented the **Minecraft Server List Ping protocol** from scratch, including VarInt encoding, TCP packet construction, handshaking and status-response decoding.
- Added configurable **timeouts, TTL-based caching, CORS, structured logging and HTTP middleware**, with unit and integration tests covering protocol and API behaviour.
- Developing a **React/TypeScript frontend** integrating live server information, an interactive world map and community features with the Go API.

NSW Crime RAG System | *Python, LangChain, ChromaDB, OpenAI, Ollama*

Oct. 2025 – Feb 2026

- Built a **retrieval-augmented generation (RAG) system** for querying NSW crime statistics, using OpenAI embeddings and ChromaDB for semantic retrieval with locally hosted Llama2 inference.
- Implemented document **chunking, vector embedding, persistent vector storage and similarity-based retrieval**, returning source documents alongside generated responses for traceability.
- Developed an **evaluation framework across 10 ground-truth queries**, measuring response effectiveness, faithfulness and source attribution to assess retrieval and generation quality.

TECHNICAL SKILLS

Languages: Go, Python, Java, C/C++, JavaScript/TypeScript, SQL

Frameworks: React, Node.js, Express, FastAPI, Chi

Infrastructure & Tools: Git, Linux, Docker, Prometheus, Neovim, AWS

Data: PostgreSQL, MS SQL Server, MongoDB, Redis